

ECO387C - Fall 2024

Discussion 9

Raymond He & Husang Kim

Investment with convex adjustment cost

The representative firm seeks to maximize:

$$\max_{\{k_{t+1}, x_t\}_{t=0}^{\infty}} \sum_{t=0}^{\infty} \left(\frac{1}{1+r} \right)^t \left(z_t k_t^\alpha - x_t - \frac{\phi}{2} x_t^2 \right)$$

subject to

$$k_{t+1} = (1 - \delta)k_t + x_t$$

and initial conditions and the non-negativity constraint for capital. Setting up the Lagrangian

$$\mathcal{L} = \sum_{t=0}^{\infty} \left(\frac{1}{1+r} \right)^t \left[\left(z_t k_t^\alpha - x_t - \frac{\phi}{2} x_t^2 \right) \right] + \sum_{t=0}^{\infty} \lambda_t ((1 - \delta)k_t + x_t - k_{t+1})$$

Note that λ_t , the present-value Lagrange multiplier, indicates the marginal value of relaxing the constraint, *discounted* in time 0 units. Define $\left(\frac{1}{1+r} \right)^t q_t = \lambda_t \Leftrightarrow q_t = (1+r)^t \lambda_t$, which is the current-value Lagrange multiplier. It shows the value to the firm of an additional unit of capital at time t in time t units. Then we can rewrite the Lagrangian as

$$\mathcal{L} = \sum_{t=0}^{\infty} \left(\frac{1}{1+r} \right)^t \left[\left(z_t k_t^\alpha - x_t - \frac{\phi}{2} x_t^2 \right) + q_t ((1 - \delta)k_t + x_t - k_{t+1}) \right]$$

Taking the first-order conditions with respect to k_{t+1} and x_t , we arrive at the following optimality conditions for the firm:

- Investment:

$$1 + \phi x_t = q_t \quad \Leftrightarrow \quad x_t = \frac{1}{\phi}(q_t - 1)$$

- Capital:

$$q_t = \frac{1}{1+r} (\alpha z_{t+1} k_{t+1}^{\alpha-1} + (1-\delta)q_{t+1})$$

- Evolution of capital

$$k_{t+1} = (1-\delta)k_t + x_t$$

In addition, the optimal solution to this problem needs to satisfy the transversality condition:

$$\lim_{\tau \rightarrow \infty} \left(\frac{1}{1+r} \right)^\tau q_\tau k_{\tau+1} = 0$$

Phase diagram

For the simplicity of analysis, assume and $z_t = z$ for all t . Then we have $\Delta k_{t+1} + \delta k_t = x_t$ and

$$\begin{cases} \Delta k_{t+1} = \frac{1}{\phi}(q_t - 1) - \delta k_t \\ \Delta q_{t+1} = \frac{r+\delta}{1-\delta}q_t - \frac{\alpha z}{1-\delta}k_{t+1}^{\alpha-1} \end{cases}$$

The first condition can be rewritten as:

$$\begin{cases} \Delta k_{t+1} > 0 & \Leftrightarrow & q_t > \phi \delta k_t + 1 \\ \Delta k_{t+1} = 0 & \Leftrightarrow & q_t = \phi \delta k_t + 1 \\ \Delta k_{t+1} < 0 & \Leftrightarrow & q_t < \phi \delta k_t + 1 \end{cases}$$

This shows that the changes in capital depend on the relative value of q_t to $\phi \delta k_t + 1$. This is illustrated in the left panel of Figure 1. In this figure, the horizontal axis represents the level of capital (k), while the vertical axis shows the value of the multiplier (q). The upward-sloping curve is the locus where $\Delta k_{t+1} = 0$. The area above and below each describes the region in which capital will increase or decrease.

Similarly, the second condition can be rewritten as:

$$\begin{cases} \Delta q_{t+1} = 0 & \Leftrightarrow & q_t = \frac{\alpha z}{r+\delta} k_{t+1}^{\alpha-1} \\ \Delta q_{t+1} > 0 & \Leftrightarrow & q_t > \frac{\alpha z}{r+\delta} k_{t+1}^{\alpha-1} \\ \Delta q_{t+1} < 0 & \Leftrightarrow & q_t < \frac{\alpha z}{r+\delta} k_{t+1}^{\alpha-1} \end{cases}$$

The second condition states that changes in the multiplier depend on the relative size of the current multiplier and marginal product of capital scaled by interest rate and depreciation. This condition can be depicted as a downward-sloping curve in the right panel of Figure 1. The area above and below each describes the region in which the multiplier will increase or decrease.

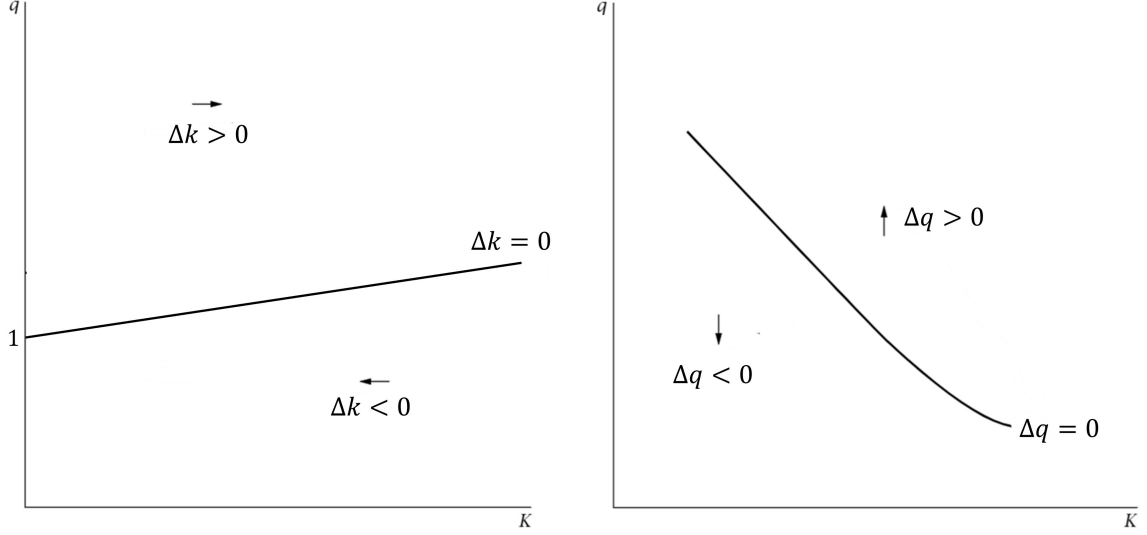


Figure 1: Dynamics of k and q

Figure 2 describes the joint dynamics of k and q under these two conditions. In each quadrant, there is a pair of arrows indicating how these two variables change. For example, a coordinate (k_0, q_0) in the lower-right quadrant will change into (k_1, q_1) such that $k_1 < k_0$ and $q_1 < q_0$. The intersection of two curves (E) identifies the steady state, where both k_t and q_t remain constant. The steady-state values $E(\bar{k}, \bar{q})$ need to satisfy the following two equations:

$$\begin{aligned}\bar{q} &= \phi\delta\bar{k} + 1 \\ \bar{q} &= \frac{\alpha z}{r + \delta}\bar{k}^{\alpha-1}\end{aligned}$$

Given these dynamics, we can analyze how the optimal capital choice evolves with the multiplier outside the steady state E. The right panel of Figure 2 shows three possible dynamics, all starting from the same initial capital k_0 but different values of q_0 :

- The first path, shown by the solid orange curve, starts from $(k_0, q_0 = \tilde{q})$. The path diverges away from the steady state and violates the non-negativity constraints.
- The second path, shown by the solid blue curve, starts from $(k_0, q_0 = \hat{q})$. The path also diverges toward infinity, resulting in an unbounded capital stock and multiplier.
- The third path in the middle, shown by the solid black curve, starts from $(k_0, q_0 = q^*)$. The path converges to the steady state (E). This path, known as the saddle path or stable arm, represents the optimal sequence of capital and multiplier that solves the firm's profit-maximization problem.

Thus, given the initial capital k_0 , it is crucial to identify the corresponding q^* which ensures that the dynamics follow the saddle path and reach the steady state.

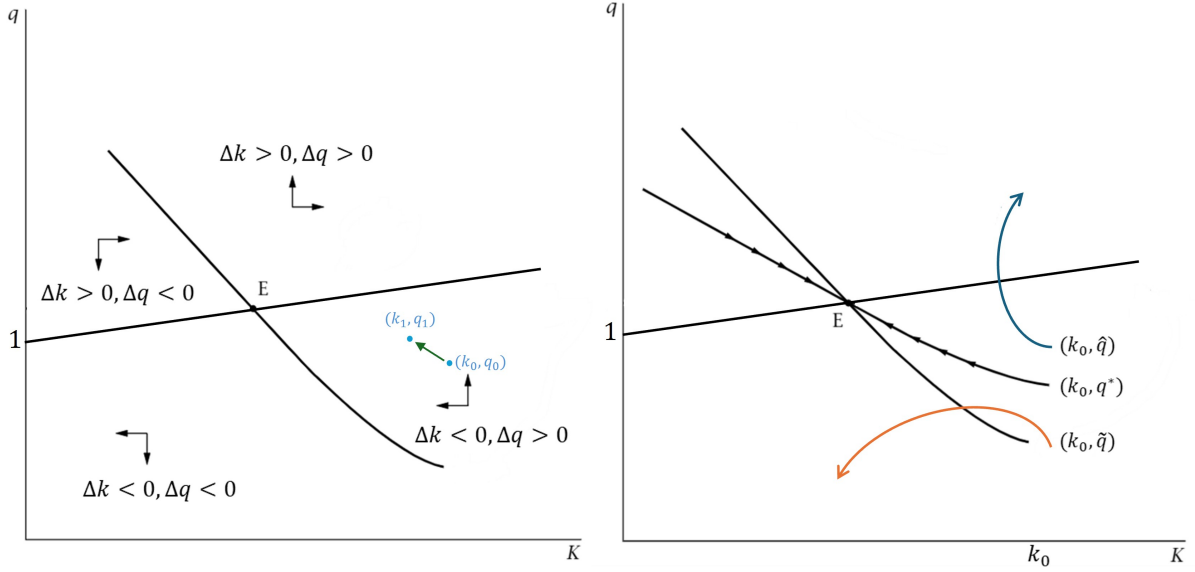


Figure 2: Dynamics of k and q , steady state, and saddle path

Transversality condition

How do we find $q_0 = q^*$? In other words, how do we remove paths like the one that begins with q_0 at $\hat{q}$ and lead to an infinite stock of capital and multiplier? The transversality condition, which prevents the overaccumulation of wealth, is the optimality condition that distinguishes the true optimal path from the explosive paths. Recall that the transversality condition in this economy is

$$\lim_{\tau \rightarrow \infty} \left(\frac{1}{1+r} \right)^\tau q_\tau k_{\tau+1} = 0$$

The saddle path that begins from (k_0, q^*) converges to a steady state $E(\bar{k}, \bar{q})$. Thus, the condition above will hold in the limit. However, the capital and multiplier on the blue path can be shown to grow at a rate greater than the interest rate, violating the transversality condition. I will heuristically demonstrate why the blue path violates the transversality condition. Recall the optimality condition for capital:

$$q_t = \frac{1}{1+r} (\alpha z_{t+1} k_{t+1}^{\alpha-1} + (1-\delta)q_{t+1})$$

We can rewrite this as

$$\frac{q_{t+1}}{q_t} = \frac{1+r}{1-\delta} - \frac{\alpha z k_{t+1}^{\alpha-1}}{(1-\delta)q_t}$$

If capital and the multiplier evolve along the explosive blue path for a sufficiently long period of time, the second term on the right-hand side will converge to zero and become negligible. In other words, for some large $t = \tau \gg 0$,

$$\frac{q_{\tau+1}}{q_\tau} \approx \frac{1+r}{1-\delta} \Rightarrow \lim_{\tau \rightarrow \infty} \left(\frac{1}{1+r} \right)^\tau q_\tau = \infty$$

This shows that in the limit, as the growth rate of the multiplier is greater than $(1+r)$, discounting with the interest rate is not sufficient enough to prevent the divergence. Then, from the optimality condition for investment, it also follows that the capital diverges as well:

$$\begin{aligned} k_{\tau+1} - (1-\delta)k_{\tau} &= \frac{1}{\phi}q_{\tau} - \frac{1}{\phi} \\ \frac{1}{(1+r)^{\tau}}k_{\tau+1} - (1-\delta)\frac{1}{(1+r)^{\tau}}k_{\tau} &= \frac{1}{\phi}\frac{1}{(1+r)^{\tau}}q_{\tau} - \frac{1}{(1+r)^{\tau}}\frac{1}{\phi} \\ \lim_{\tau \rightarrow \infty} \left\{ \frac{1}{(1+r)^{\tau}}k_{\tau+1} - (1-\delta)\frac{1}{(1+r)^{\tau}}k_{\tau} \right\} &= \lim_{\tau \rightarrow \infty} \left\{ \frac{1}{\phi}\frac{1}{(1+r)^{\tau}}q_{\tau} - \frac{1}{(1+r)^{\tau}}\frac{1}{\phi} \right\} \end{aligned}$$

Since the right-hand side diverges, the left-hand side also diverges as well. This implies that $\lim_{\tau \rightarrow \infty} \frac{1}{(1+r)^{\tau}}k_{\tau+1} = \infty$ and the violation of the transversality condition immediately follows.

$$\lim_{\tau \rightarrow \infty} \left(\frac{1}{1+r} \right)^{\tau} q_{\tau} k_{\tau+1} = \infty \neq 0$$

Thus, the blue path associated with the inefficient accumulation of capital cannot be the optimal solution to the firm's problem because it violates the transversality condition.

Example

Forward-iteration of the optimality condition for capital at time 0 gives:

$$\begin{aligned} q_0 &= \frac{1}{1+r} (\alpha z k_1^{\alpha-1} + (1-\delta)q_1) \\ &= \frac{1}{1+r} \left(\alpha z k_1^{\alpha-1} + \frac{1-\delta}{1+r} (\alpha z k_2^{\alpha-1} + (1-\delta)q_2) \right) \\ &= \vdots \\ &= \frac{1}{1-\delta} \sum_{t=1}^{\infty} \left(\frac{1-\delta}{1+r} \right)^t \alpha z k_t^{\alpha-1} + \lim_{\tau \rightarrow \infty} \left(\frac{1-\delta}{1+r} \right)^{\tau} q_{\tau} \end{aligned}$$

Note that the transversality condition implies:

$$\lim_{\tau \rightarrow \infty} \left(\frac{1}{1+r} \right)^{\tau} q_{\tau} k_{\tau+1} = 0 \quad \Rightarrow \quad \lim_{\tau \rightarrow \infty} \left(\frac{1}{1+r} \right)^{\tau} q_{\tau} = 0 \quad \Rightarrow \quad \lim_{\tau \rightarrow \infty} \left(\frac{1-\delta}{1+r} \right)^{\tau} q_{\tau} = 0$$

The first relationship follows as capital and the multiplier are tightly connected through the optimality conditions and approach the constant steady state on the stable arm. The second relationship follows from the fact that $\frac{1-\delta}{1+r} < \frac{1}{1+r}$. Thus, imposing the transversality condition gives a unique q_0 :

$$q_0^* = \frac{1}{1-\delta} \sum_{t=1}^{\infty} \left(\frac{1-\delta}{1+r} \right)^t \alpha z k_t^{*\alpha-1}$$

This is the q_0^* , combined with initial capital k_0 , which constitutes the initial point on the saddle path that respects all the optimality conditions (including TVC) and constraints. This condition says that if the path of capital $\{k_{t+1}^*\}_{t=0}^\infty$ is chosen such that it respects the FONCs, and transversality condition, then the q_0 will be the present discounted value of the marginal product of capital.

For example, if the initial level of capital was at the steady state $k_0 = \bar{k}$, the optimal sequence of capital that respects all the optimality conditions and constraints will also be the steady state capital: $k_{t+1} = \bar{k}$, for all t . Then, the initial level of multiplier q_0 that is consistent with this choice is

$$q_0^* = \frac{1}{1-\delta} \sum_{t=1}^{\infty} \left(\frac{1-\delta}{1+r} \right)^t \alpha z \bar{k}^{\alpha-1} = \frac{\alpha z}{r+\delta} \bar{k}^{\alpha-1} = \bar{q}$$

which is the steady state multiplier. This means that given initial capital of k_0 , the optimal sequence of capital and investment for the firm is

$$k_{t+1} = \bar{k}, \quad x_t = \delta \bar{k} \quad \text{for all } t$$

This is the only sequence that respects all the FONCs, transversality condition, and constraints.